LAB 03 SOLUTIONS:

Exercise 01:

Perform the given operations

I. a Python program to square and cube every number in a given list of integers using Lambda.

CODE:

```
33  # EXERCISE 01
34  # === i ===
35  sq = lambda a: a*a
36  cube= lambda x : x*x*x
37
38  numbers= [1,2,3,4,5]
39  for n in numbers:
40      print("Number : ",n," it's square : ", sq(n), " and cube : ",cube(n))
41
```

OUTPUT:

```
Number :  1  it's square :  1  and cube :  1
Number :  2  it's square :  4  and cube :  8
Number :  3  it's square :  9  and cube :  27
Number :  4  it's square :  16  and cube :  64
Number :  5  it's square :  25  and cube :  125
PS C:\Users\hp\Desktop\UBIT\SEMESTER 4\AI\uni labs\lab03>
```

II. a Python program to find if a given string starts with a given character using Lambda.

CODE:

```
ans= lambda string,char : "yes" if string[0].lower() ==char.lower() else "no"
print(ans("Software","s"))
```

OUTPUT:

```
yes
PS C:\Users\hp\Desktop\UBIT\SEMESTER 4\AI\uni_labs\lab03>
```

III. a Python program to extract year, month, date and time using Lambda.

CODE:

```
from datetime import datetime

year= lambda x: x.year
month= lambda x: x.month
date= lambda x: x.day
time= lambda x: x.time()

d= datetime.now() #for today's date

print("Year : ",year(d))
print("Month : ",month(d))
print("Date : ",date(d))
print("Time : ",time(d))
```

OUTPUT:

```
Year :  2026
Month :  9
Date :  7
Time :  12:57:47.171041
PS C:\Users\hp\Desktop\UBIT\SEMESTER 4\AI\uni_labs\lab03>
```